Study Memo (Condensed): Does Machine Learning Beat the Credit Scorecard?

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Concepts in one line each

PD: probability a loan applicant defaults, estimated from origination-time information only.

Charged off / fully paid: the two resolved outcomes; loans still in flight are excluded so the target is fully observed.

Leakage: using post-origination fields (payments, recoveries); cured by a whitelist of 23 origination-time columns out of 151.

Right-censoring: unfinished loans have unknown outcomes; avoided by using only 36-month loans issued 2007–2015.

Out-of-time split: train ≤2013, calibrate 2014, test 2015; reproduces deployment and keeps the default-rate drift (12.6 → 14.9%).

WoE: per bin, $\ln[(g_j/g)/(b_j/b)]$; puts every feature on one evidence scale, handles missing values, keeps the model readable.

IV: $\sum_j (g_j/g - b_j/b)\mathrm{WoE}_j$; screening keeps features with IV ≥ 0.02 (11 of 23 kept).

Scorecard: logistic regression on WoE features, rescaled so 600 points = 19:1 odds and +20 points doubles the odds.

Boosted trees: XGBoost / LightGBM, sequential shallow trees fit to loss gradients, early-stopped on validation AUC.

Shared design matrix: identical inputs for all challengers (median-impute + one-hot, fit on train), so gaps come from the learner.

AUC: probability a random defaulter outranks a random good borrower; ranking only. **KS**: max gap between score CDFs.

Calibration: among loans with $\hat{p} \approx q$, a fraction q defaults. **Brier**: MSE of the probability. **ECE**: mean |predicted − observed| over 10 bins. **Reliability diagram**: the picture of it.

Platt / isotonic: monotone recalibration maps (logistic on the logit / step function), fit on the 2014 vintage, applied to 2015, same for every model; monotone maps leave AUC unchanged.

DeLong test: significance of an AUC difference when both models score the same test set (correlated AUCs).

Bootstrap CI: percentile intervals from 1,000 resamples of the test set.

Profit model: accept if $\hat{p} \le c$, sweep c; repaid loan earns 36·installment – principal, charged-off loses 0.65×principal; report USD per 1,000 applicants; curve maximum = best operating point.

Reject inference: the unsolved problem that only accepted loans are observed; stated as a limitation.

Results in five lines

XGBoost 0.6850 AUC vs scorecard 0.6767 (+0.85 pts, $p < 10^{-54}$); LightGBM +0.72; MLP –0.17, significantly worse.

All raw models under-predict PD (2015 defaulted more than training years); isotonic fixes most, not all.

Scorecard stays best-calibrated (ECE 0.0175 vs 0.0214); trees keep the better Brier.

Profit: accept-all $705,171 per 1,000; scorecard best +1.2%; XGBoost best +3.3% (+$14,700 over scorecard, +2.1%).

Taiwan robustness reproduces the ranking (+1.50 pts, $p = 2.8 \times 10^{-6}$). Verdict: a qualified yes, trees only.

Quick-fire Q&A

Q. Why not judge by AUC alone? A. AUC sees only ranking; provisioning needs the PD values right (calibration) and lending needs money made at a cutoff (profit). Raw trees had the best AUC and the worst ECE simultaneously.

Q. One concrete leaky feature? A. Principal received to date: equals loan amount for repaid loans, so it encodes the answer.

Q. Why out-of-time, not cross-validation? A. Deployment scores future applicants under a different economy; random splits share the regime across train and test, flattering flexible models and hiding drift.

Q. Why do banks still use scorecards? A. Stable, auditable, point-readable, regulator-friendly; and here, best-calibrated with most of the trees' performance.

Q. Why fit calibrators on 2014? A. Must be out-of-sample for the model and recent for the deployment; identical treatment keeps the comparison fair.

Q. Why is calibration still imperfect after isotonic? A. The map is learned on 2014; 2015's higher default level was not yet visible. Monotone maps cannot anticipate the future, hence ongoing monitoring in practice.

Q. Tiny p-values: is the edge big? A. No; 283k test loans make any real gap significant. Size is the point: +0.85 AUC pts, +2.1% profit.

Q. Why did the MLP lose? A. Tabular data favours trees' and binning's inductive biases; the MLP adds flexibility the data cannot pay for, with 10× the seed variance.

Q. Why is accept-everyone profitable, and so what? A. Interest income exceeds default losses; the model only adds value pruning the riskiest tail, so models separate only near the optimal cutoff.

Q. Isn't keeping grade/int_rate circular? A. It is the investor's information set; it compresses the gap (all models re-rank ranked loans) and is stated as a limitation. Raw bureau data would likely widen the ML edge.

Q. Biggest lesson? A. Evaluation design beats modelling: the whitelist, the calendar split and the choice of criteria changed conclusions more than any hyperparameter; "best model" depends on the question.

Code and results: `https://github.com/iitgF/T8`. *Full walkthrough:* `memo_standard.tex`.